

Ray Foysal

+353 85 103 1313 +61 432 743 233 | rayfoysal@gmail.com | [LinkedIn: Ray Foysal](#) | [Project Portfolio](#)

Education

Australian National University (ANU) | Canberra, Australia **2025 - Present**

- Completing my 3rd year of mechanical engineering at ANU as part of an Exchange Programme

University College Dublin (UCD) | Dublin, Ireland **2023 - Present**

- Grade 1:1
- Relevant Modules: Mechanics for Engineers, Mechanics of Fluids, Manufacturing Engineering, Energy Engineering, Physics for Engineers, Computer Science for Engineers, Computer Engineering, Robotics Design Project, Electrical and Electronic Circuits, Creativity in Design.

Colaiste Pobail Setanta | Dublin, Ireland **2018 - 2023**

- Leaving Certificate Points Obtained: **613**

Experience

UCD UAV | Aeromechanical & Simulations Engineer **Feb 2024 – Jul 2025**

- Utilized X-Plane 11 simulations to refine flight characteristics of different planes in our prototype series.
- Used SimScale to simulate and verify properties of potential landing gear components to select the best option.
- Collaborated with the Avionics department to design a transport container for the UAV while innovating by implementing the ground control station to maximise ease of transportation.
- Conducted analysis of air foil designs and lift/drag properties using XFLR5 CFD software.
- Competed in Buckminster UK to win two awards for business presentation and airworthiness, while achieving runner-up in 4 categories out of 8 total against 32 international teams.

UCD Formula Student | Powertrain Engineer **Sep 2023 - Present**

- Contributed to the development of the High Voltage Battery Enclosure by designing it with SolidWorks and Onshape.
- Evaluated appropriate high voltage fuses for use in internal battery connections, and created technical reports for review by faculty advisors.
- Used KiCAD to reference circuit board schematics in order to solder components correctly.
- Involved in creation of the car's fiberglass hood during production to reduce car's weight by 10%.
- Assembled wheel connections such as tie rods, knuckles, axle, and suspension components.
- Participated in the IMechE FSUK 2024 competition as a member of the UCDFS team and secured 39th position at Silverstone.

Work Experience

Red Cow Moran Hotel | F&B Assistant **Sep 2023 – April 2025**

- Received Manual Handling and HACCP training.
- Worked in a fast paced environment, served 100+ people every day.

Talbot Hotel Stillorgan | F&B Assistant / Receptionist **February – March 2022**

- Assisted in daily operations, managed guest inquiries, and helped to ensure a high level of 95% customer satisfaction.

Technical Skills

Programming

- Proficiency of C/C++ Programming from Computer Science, Computer Engineering and Robotics modules undertaken at UCD.

- Knowledge of Python from Programming for Scientists, and Environmental Sensing and Mapping modules taken at ANU.

Modelling/Simulating

- Proficiency in SOLIDWORKS, OnShape and Autodesk Fusion 360
- Knowledge of MATLAB and Simulink from Engineering Computing course undertaken in UCD, and from battery simulation trials for the formula student team in UCD.
- Knowledge of Blender 3D modelling software to create several animations for Engineers Without Borders video presentation and for product advertisements for personal e-commerce business.
- Knowledge of Simulating softwares such as SimScale, XFLR5, and X-Plane 11 from experience working on the UCD UAV team.

Manufacturing

- Knowledge in 3D printing, manual machining and CNC machining.
- Knowledge of materials and associated properties; thermoset/thermoplastic polymers, composites, metals etc.

Circuitry & Electronics

- Knowledge in LTSpice simulation and KiCAD PCB Modelling.
- Knowledge in Arduino, sensors, shields, and circuitry from multiple projects; RoboWars, Robotics Design Module, MicroMouse Competition, NASA SpaceApps Hackathon.

Microsoft Suite

- Proficiency in MS Excel from specific coursework done in Intro to Eng. Computing, along with MS Word and PowerPoint.

Achievements/Extra Curricular

Dogpatch RoboWars 2025 | Winner

July 2025

- Competed in RoboWars battle organised by Dogpatch, a startup and innovation hub in Dublin's docklands known as the city's "Silicon Docks".
- Worked in a team of 6 people to create a robot from scratch to fight in the timeframe of 1 month.

NASA Space Apps Hackathon | Winner

October 2023

- Achieved first place in the largest hackathon in Ireland, using NASA's open data.
- Responsible for designing and 3d printing the prototype beacon that detects forest fires from its initial stages and warns local authorities.

Produced AI Podcast

February 2024

- Served as the producer for a podcast episode on AI and its impact on reflective assignments as part of a series from the UCD College of Engineering and Architecture.
- Collaborated with the facilitators, team members and guests to plan, script and record the episode.
- Coordinated logistics, including managing recording equipment, booking recording times at a studio and post-production editing to ensure high quality content delivery.

UCD Entrance Scholar

March 2024

- Awarded UCD Entrance Scholar Award for achieving outstanding result in the Leaving Certificate and graduating as the top student in secondary school.

Robotics Design Competition | Software Developer and Design Engineer

May 2024

- Developed and implemented a strategy in C programming for the robot, resulting in a top-half finish in the competition.
- Designed mechanical systems within the robot to combat and outlast opponents.

Colaiste Pobail Setanta | Best Senior Certificate

June 2024

- Awarded to the senior from secondary school with the highest leaving certificate result.

Sports Star of The Year

June 2021

- Recognised as the best performing student athlete in secondary school.